(c) North (d) Sky

47th B.P.S.C. (Pre) 2005

Ans. (c)

The magnetic needle points to north. All magnets have two poles, north pole and south pole. The north pole of one magnet is attracted to the south pole of the another magnet. The earth is behave like a magnet that can interact with other magnets in this way. So the north end of a compass magnet is drawn to align with the earth's magnetic field. Because the earth's magnetic north pole attracts the 'north' ends of other magnets, it is technically the 'south pole' of our planet's magnetic field.

9. What was the fissionable material used in the bombs dropped at Nagasaki (Japan) in the year 1945?

(a) Sodium

(b) Potassium

(c) Plutonium

(d) Uranium

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

'Fat Man' was the code name for the type of bomb which was dropped on the Japanese city of Nagasaki on 9 August, 1945 by the United States of America. Plutonium was used as fissionable material in this bomb. On the other hand 'Little Boy' which was dropped on the Japanese city of Hiroshima on August 6, 1945 used Uranium as fissionable material.

10. Solar energy is obtained from -

(a) Moon (b) sea

(c) Sun (d) air

44th B.P.S.C. (Pre) 2000-01

Ans. (c)

Solar energy is obtained from the Sun. Due to the nuclear fusion in the core of the Sun, a huge amount of energy is released continuously, i.e., energy is generated on the Sun by nuclear fusion. This energy continues to radiate in all directions in space. The Earth and other solar planets get only a small part of this energy. The Earth receives only about 4.5×10^{-8} parts of energy.

11. Which of the following Nobel gas is not present in air?

(a) Helium (b) Neon

(c) Organ (d) Radon

46th B.P.S.C. (Pre) 2004

Ans. (d)

Radon inert gas is not found in air, other inert gases helium, neon and argon are present in air.

12. During thunderstorm, it is safer to

(a) carry an open umbrella

(b) take shelter under short tree

(c) stand in open field

(d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

During thunderstorm, it is safer to take shelter under short tree among carry on open umbrella, stand in open field and take shelter under short tree. Lighting most get attracted by tall targets.

13. What is the full form of TRP used in TV channels?

(a) Television Reverse Point (b) Television Result Point

(c) Telecast Reverse Point (d) Television Rating Point

(e) None of the above / More than one of the above

B.P.S.C. Project Manager Prelims 2021

Ans. (d)

The full form of TRP is Television Rating Point or Target Rating Point. TRP is a metric that signifies a programme's success on tv screens.

14. Which is the first State of India to use robotic technology to clean all its commissioned manholes?

(a) Bihar (b) Punjab

(c) Kerala (d) Tamil Nadu

BPSC Agriculture Department-2024

Ans. (c)

Kerala is the first State of India to use robotic technology to clean all its commissioned manholes. The Kerala Water Authority (KWA) launched the robotic scavenger, Bandicoot. The robot was developed by Genrobotics, a Kerala-based company.

II. CHEMISTRY

Atomic Structure

1. Who is regarded as the Father of Modern Chemistry?

(a) Rutherford (b) Einstein

(c) Lavoisier (d) C.V. Raman

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

French nobleman and Chemist Antoine Lavoisier is regarded as the 'Father of Modern Chemistry'. Lavoisier is most noted for his discovery of the role of oxygen plays in combustion. He recognized and named oxygen (1778) and hydrogen (1783), wrote the first extensive list of elements and helped to reform chemical nomenclature.

2. **The positively charged part at the centre of an atom is called as :**
(a) Proton (b) Neutron
(c) Electron (d) Nucleus
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

The positively charged part at the centre of an atom is called as nucleus. The atomic nucleus is the small, dense region consisting of proton and neutron at the centre of an atom.

3. **Constituents of atomic nucleus are –**
(a) Electron and proton
(b) Electron and neutron
(c) Proton and neutron
(d) Proton, neutron and electron

41st B.P.S.C. (Pre) 1996
43rd B.P.S.C. (Pre) 1999

Ans. (c)

The constituents of an atom are protons, neutrons and electrons. The protons and neutrons (nucleons) are found in the nucleus of atoms. The nucleus of an atom is surrounded by electrons.

4. **A single type of atom is found in –**
(a) Compounds of minerals
(b) Mixture of minerals
(c) Native elements
(d) None of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

Native element is a material that consists of a single type of atom, while a compound element consists of two or more types of atoms.

5. **'God particle' is :**
(a) Neutrino (b) Higgs Boson
(c) Meson (d) Positron
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

The Higgs Boson is an elementary particle in the standard model of particle physics. The hypothesis of its existence was given in 1964 but practically it was proved on March 14, 2013. In mainstream media the Higgs Boson has often been called the 'God particle'. The two physicists who discovered these particles Peter W. Higgs and Francois Englert were awarded the Nobel Prize in Physics in 2013.

6. **The radius of the n-th stationary orbit of hydrogen atom is proportional to**
(a) $\frac{1}{n^2}$ (b) $\frac{1}{n}$
(c) n^2 (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 6-8) 10-12-2023

Ans. (c)

The radius of the nth stationary orbit of hydrogen atom is proportional to n^2.

7. **Which particle is free of charge?**
(a) α-particle (b) Electron
(c) Neutron (d) Proton
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

Among the given particles neutron is free of charge. Alpha (α) particle (+2ve) and proton (+ve) are positive charged particles while electron (–ve) carries negative charge.

8. **Which of the following elements does not contain neutrons?**
(a) Oxygen (b) Nitrogen
(c) Hydrogen (e) Copper
(d) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (c)

The atomic no. of Hydrogen is 1. The Protium ($_1H^1$) isotope of the Hydrogen has no neutron while Deuterium ($_1H^2$) has one neutron and Tritium ($_1H^3$) isotope has 2 neutrons.

9. **Which of the following carries a negative charge?**
(a) X-rays (b) Alpha particles
(c) Beta particles (d) Gamma rays

45th B.P.S.C. (Pre) 2002

Ans. (c)

Alpha rays or Alpha particles (α) are the positively charged particles. Beta particles (β) are highly energetic electrons which are released from inner part of a nucleus. They are negatively charged (-1e) and have a negligible mass. Gamma radiation (γ) consist of photons, which travel at the speed of light like all electromagnetic radiations. X-ray has no mass or charge. Gamma radiation can travel much faster (speed of light) in the air than alpha and beta.

10. **A photon is**
(a) a quanta of light energy
(b) a quanta of material (substance)

(c) a positively charged particle
(d) a light intensity gauge
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (e)

A photon is an elementary particle that is a quantum of the electromagnetic field, including electromagnetic radiation such as light and radio waves, and the force carrier for the electromagnetic force. Photons are massless, so they always move at the speed of light in vacuum when they are in vacuum.

11. The speed of an electron in the orbit of the hydrogen atom in the ground state is
(a) c
(b) c/2
(c) c/137
(d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 08-12-2023

Ans. (c)

The speed of an electron in the orbit of the hydrogen atom in the ground state is c/137 where c is the speed of light. The value of c is 3×10^8 m/s.

12. The number of electrons and neutrons in an element is 18 and 20 respectively. Its mass number is :
(a) 22 (b) 2
(c) 38 (d) 20
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Given that,
number of neutrons = 20
& number of electrons = 18
Thus, number of protons = number of electrons = 18
Mass number = number of protons + number of neutrons
= 18 + 20 = 38

13. What is the mass number of an element, the atom of which contains two protons, two neutrons and two electrons?
(a) 2 (b) 4
(c) 6 (d) 8

43rd B.P.S.C. (Pre) 1999

Ans. (b)

The mass number of an element is the sum of a total number of protons and neutrons inside in its nucleus and represents by A. Therefore, mass number (A) = number of protons + number of neutrons.
Therefore, mass number (A) = 2 + 2 = 4

14. The mass number of a nucleus is -
(a) the sum of the numbers of neutrons and protons
(b) the total mass of neutrons and protons
(c) always greater than the atomic mass
(d) always less than its atomic number
(e) none of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (a)

See the explanation of the above question.

15. The number of neutrons in the nucleus of plutonium nuclide ($_{94}Pu^{242}$) is :
(a) 94 (b) 148
(c) 242 (d) 336
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

In $_{94}Pu^{242}$,
Number of protons = 94, and
Number of protons plus neutrons = 242
The number of neutrons in the nucleus of plutonium nuclide ($_{94}Pu^{242}$) is 242 – 94 = 148.

16. Particles which can be added to the nucleus of an atom without changing its chemical properties are
(a) neutrons (b) electrons
(c) protons (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 11-12) 15-12-2023

Ans. (a)

Neutrons, as they are neutral particles and can be added to the nucleus of an atom without changing its chemical properties .

17. Isotopes are that nuclei of atomic nucleus in which –
(a) Number of neutrons is same but number of protons is different
(b) Number of protons is same but number of neutrons is different
(c) Number of both protons and neutrons is same
(d) Number of both protons and neutrons is different

41st B.P.S.C. (Pre) 1996

Ans. (b)

The Isotopes are a set of nuclides/atoms having the same number of protons, but a different number of neutrons. In other words, the same atomic number but having different atomic mass. Each individual isotope has separate nuclei. The Isotopes that are unstable and undergo radioactive decay are called radioisotopes.